

Replication and Extension of Jealousy Responses to Rival Characteristics

Applied Stats II: Replication Project by Whiz

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Abstract

This report replicates a 2020 evolutionary psychology study on jealousy responses to romantic rivals and extends the analysis in two ways. The original study examined whether men and women respond differently to rival attractiveness and rival dominance, with particular attention to the interaction between rival attractiveness and participant gender. Using the authors' publicly available data and code structure, I reproduced the main jealousy models for Study 1 and Study 2.

I then implemented two extensions. First, I conducted a direct cross-study robustness comparison of the *Attractive condition* $\times$ *Gender* interaction, estimating the same model in each study and then comparing the effect across samples. Second, I re-estimated the core relationship using an ordered logit model, which is more appropriate for an ordinal jealousy measure and aligns with the course material on generalized linear models and ordered choice models.

The results show that the attractiveness-by-gender pattern is strongest in Study 1 and weaker in Study 2. The ordered logit analysis broadly supports the same substantive interpretation, but also suggests that the magnitude and stability of the effect are more limited than a single-study reading would imply. Overall, the replication confirms the original study's main pattern only partially and shows that its central result is sensitive to sample and model specification.

1 Introduction

Jealousy is a central topic in evolutionary psychology because it is often interpreted as an adaptive response to threats to romantic relationships. One influential line of argument is that men and women may respond differently to different types of romantic rivals. In particular, the original study revisited a classic claim that women should be especially sensitive to rival attractiveness, whereas men should be especially sensitive to rival dominance. My project focuses on the jealousy outcome and the study’s central empirical question: whether the effect of rival attractiveness on jealousy differs by gender.

This project has two goals. First, it provides an exact replication of the main jealousy analysis from the original study. Second, it extends that analysis using methods closely related to course material. The first extension is a robustness comparison across the paper’s two replication studies. The second is a model-based extension that re-estimates the focal relationship using ordered logit, which is appropriate for ordinal outcomes and reflects the course emphasis on generalized linear models, binary and ordered choice models, and model interpretation.

2 The Original Study

The original study examines whether jealousy differs depending on two rival characteristics: attractiveness and dominance. The key theoretical claim is that women may respond more strongly to physically attractive rivals, while men may respond more strongly to dominant rivals. The study includes two replication samples and estimates factorial models in which jealousy is the dependent variable and rival attractiveness, rival dominance, and participant gender are the predictors.

The replication script reconstructs this design directly. For both Study 1 and Study 2, the script reads online and field datasets, restricts the sample to respondents currently in a relationship, constructs the *Attractive condition* and *Dominant condition* variables from the vignette indicators, and estimates a three-way factorial ANOVA using `ezANOVA()`. The main original specification is therefore a model of jealousy as a function of attractiveness, dominance, gender, and all interactions among them.

3 Data and Empirical Strategy

The data consist of two replication studies, combining online and field samples. After restricting to participants currently in a relationship and performing the necessary cleaning, the datasets used in this report contain approximately 339 observations in Study 1 and 456 in Study 2. These cleaned datasets were constructed by combining subsamples and creating the experimental condition variables for rival attractiveness and rival dominance.

The empirical strategy unfolds in three stages:

1. **Replication:** Reproduce the original factorial ANOVA results for each study, focusing on the core interaction between rival attractiveness and participant gender.
2. **Cross-study comparison:** Estimate the same linear model separately in Study 1 and Study 2, extract the key interaction term, and examine whether the effect differs across samples. This formalizes robustness testing rather than simply presenting two separate analyses.
3. **Ordered logit re-analysis:** Treat jealousy as an ordinal outcome and estimate ordered logit models using `polr()`. Compare these results with the linear model findings to assess the degree to which conclusions depend on modeling choices appropriate for ordinal data.

Throughout, the analysis emphasizes the Attractive condition $\times$ Gender interaction, as it captures the central theoretical claim.

4 Replication Results

The exact replication reproduces the main jealousy ANOVA for each study. Consistent with the original focus, the interaction of interest is Attractive condition $\times$ Gender.

- **Study 1:** The interaction is evident and comparatively strong. The pattern suggests that rival attractiveness has a larger impact on women’s jealousy than on men’s.
- **Study 2:** The interaction appears much weaker. Women’s jealousy increases only slightly when the rival is attractive, while men’s jealousy shows little change or even decreases slightly. The gender difference observed in Study 1 is therefore not mirrored in Study 2.

Table 1: ANOVA Results (Study 1)

Effect	DFn	DFd	SSn	SSd	F	p	pi.05	ges
(Intercept)	1	331	254.738	509.486	165.497	5.43×10^{-31}	*	0.3333
Attractive_condition	1	331	0.337	509.486	0.219	0.640		0.0007
Dominant_condition	1	331	4.881	509.486	3.171	0.076		0.0095
Gender	1	331	1.281	509.486	0.832	0.362		0.0025
Attractive_condition:Dominant_condition	1	331	0.405	509.486	0.263	0.609		0.0008
Attractive_condition:Gender	1	331	10.080	509.486	6.549	0.011	*	0.0194
Dominant_condition:Gender	1	331	2.219	509.486	1.442	0.231		0.0043
Attractive_condition:Dominant_condition:Gender	1	331	0.056	509.486	0.037	0.849		0.0001

Table 2: ANOVA Results (Study 2)

Effect	DFn	DFd	SSn	SSd	F	p	pi.05	ges
(Intercept)	1	448	349.618	691.647	226.458	1.03×10^{-41}	*	0.3358
Attractive_condition	1	448	1.535	691.647	0.994	0.319		0.0022
Dominant_condition	1	448	0.816	691.647	0.528	0.468		0.0012
Gender	1	448	1.721	691.647	1.115	0.292		0.0025
Attractive_condition:Dominant_condition	1	448	0.120	691.647	0.078	0.781		0.0002
Attractive_condition:Gender	1	448	1.901	691.647	1.231	0.268		0.0027
Dominant_condition:Gender	1	448	0.239	691.647	0.154	0.694		0.0003
Attractive_condition:Dominant_condition:Gender	1	448	0.645	691.647	0.418	0.518		0.0009

These replication findings imply that the original study’s central effect is partially reproducible—clear in one sample but attenuated in another. This observation already cautions against overgeneralizing results from a single study.

5 Replication Extensions

5.1 Extension 1: Cross-Study Robustness Comparison

The first extension evaluates whether the core attractiveness-by-gender pattern is robust across the two replication studies. Instead of treating the studies separately, this extension directly compares the key interaction across both datasets. This approach reframes the original study's two-study design as a robustness exercise rather than simply two independent replications.

```
1 print(table(combined$Dominant_condition))
2
3 # -----
4 # 4. Same model in each study
5 # -----
6 model_s1 <- lm(
7   Jealousy ~ Attractive_condition * Dominant_condition * Gender,
8   data = s1
9 )
10
11 model_s2 <- lm(
12   Jealousy ~ Attractive_condition * Dominant_condition * Gender,
13   data = s2
14 )
15
16 anova_s1 <- anova(model_s1)
17 anova_s2 <- anova(model_s2)
18
19 print(anova_s1)
20 print(anova_s2)
21
22 # -----
23 # 5. Extract the key interaction
24 # -----
25 get_term_row <- function(anova_obj, possible_terms, study_label) {
26   rn <- rownames(anova_obj)
27   hit <- rn[rn %in% possible_terms]
28
29   if (length(hit) == 0) {
```

```

30     stop(
31         paste0(
32             "Could not find the target interaction in ", study_label,
33             ".\nAvailable rows:\n",
34             paste(rn, collapse = "\n")
35         )
36     )
37
38     term <- hit[1]
39     idx <- which(rn == term)
40
41     data.frame(
42         Study = study_label,
43         Term = term,
44         Df = anova_obj[idx, "Df"],
45         Sum_Sq = anova_obj[idx, "Sum Sq"],
46         Mean_Sq = anova_obj[idx, "Mean Sq"],
47         F_value = anova_obj[idx, "F value"],
48         p_value = anova_obj[idx, "Pr(>F)"],
49         row.names = NULL
50     )
51 }
52
53 interaction_terms <- c(
54     "Attractive_condition:Gender",
55     "Gender:Attractive_condition"
56 )
57
58 interaction_s1 <- get_term_row(anova_s1, interaction_terms, "Study
59     1")
60
61 interaction_s2 <- get_term_row(anova_s2, interaction_terms, "Study
62     2")
63
64 interaction_results <- bind_rows(interaction_s1, interaction_s2)
65 print(interaction_results)
66
67 write.csv(interaction_results, "interaction_results_by_study.csv",

```

```

    row.names = FALSE)
65
66 # -----
67 # 6. Summary table for plot and slides
68 # -----
69 summary_table <- combined %>%
70   group_by(Study, Gender, Attractive_condition) %>%
71   summarise(
72     N = n(),
73     Mean_jealousy = mean(Jealousy, na.rm = TRUE),
74     SD_jealousy = sd(Jealousy, na.rm = TRUE),
75     SE_jealousy = SD_jealousy / sqrt(N),
76     CI_low = Mean_jealousy - 1.96 * SE_jealousy,
77     CI_high = Mean_jealousy + 1.96 * SE_jealousy,
78     .groups = "drop"
79   )
80
81 print(summary_table)
82 write.csv(summary_table, "jealousy_summary_table.csv", row.names =
    FALSE)
83
84 # -----
85 # 7. Main extension plot
86 # -----
87 p_main <- ggplot(
88   summary_table,
89   aes(
90     x = Attractive_condition,
91     y = Mean_jealousy,
92     group = Gender,
93     shape = Gender,
94     linetype = Gender
95   )
96 ) +
97   geom_line(position = position_dodge(width = 0.10)) +
98   geom_point(size = 3, position = position_dodge(width = 0.10)) +
99   geom_errorbar(
100     aes(ymin = CI_low, ymax = CI_high),

```

```

101     width = 0.08,
102     position = position_dodge(width = 0.10)
103 ) +
104 facet_wrap(~ Study) +
105 labs(
106     title = "Jealousy by rival attractiveness, gender, and study",
107     x = "Rival attractiveness condition",
108     y = "Mean jealousy"
109 ) +
110 theme_minimal(base_size = 12)
111
112 print(p_main)
113
114 ggsave(
115     filename = "extension_plot_main.png",
116     plot = p_main,
117     width = 9,
118     height = 5,
119     dpi = 300
120 )

```

5.2 Extension 2: Ordered Logit Re-analysis

The second extension re-estimates the focal relationship using ordered logit. This model is more appropriate for an ordinal jealousy variable than a standard linear model. It allows for a more accurate representation of the outcome variable and provides a robustness check on the original findings.

```

1 write.csv(ord_interaction_results,
2           "ordered_logit_interaction_results.csv", row.names = FALSE)
3
4 # -----
5 # 6. OLS comparison model
6 # Same focal specification for comparison
7 # -----
8 ols_model_s1_simple <- lm(
9     Jealousy_num ~ Attractive_condition * Gender,
10    data = s1_ord
11 )

```



```

11
12 ols_model_s2_simple <- lm(
13   Jealousy_num ~ Attractive_condition * Gender,
14   data = s2_ord
15 )
16
17 ols_anova_s1_simple <- anova(ols_model_s1_simple)
18 ols_anova_s2_simple <- anova(ols_model_s2_simple)
19
20 print(ols_anova_s1_simple)
21 print(ols_anova_s2_simple)
22
23 write.csv(as.data.frame(ols_anova_s1_simple),
24           "ols_simple_study1.csv", row.names = TRUE)
25 write.csv(as.data.frame(ols_anova_s2_simple),
26           "ols_simple_study2.csv", row.names = TRUE)
27
28 # Pull out the same interaction from OLS ANOVA
29 get_ols_interaction <- function(anova_obj, study_label) {
30   rn <- rownames(anova_obj)
31   idx <- which(rn == "Attractive_condition:Gender")
32
33   if (length(idx) == 0) {
34     stop(
35       paste0(
36         "Could not find OLS interaction term in ", study_label,
37         ".\nAvailable rows:\n",
38         paste(rn, collapse = "\n")
39       )
40     )
41   }
42
43   data.frame(
44     Study = study_label,
45     Term = rn[idx],
46     Df = anova_obj[idx, "Df"],
47     Sum_Sq = anova_obj[idx, "Sum Sq"],
48     Mean_Sq = anova_obj[idx, "Mean Sq"],

```

```

46     F_value = anova_obj[idx, "F value"],
47     p_value = anova_obj[idx, "Pr(>F)"],
48     row.names = NULL
49 )
50 }
51
52 ols_interaction_s1 <- get_ols_interaction(ols_anova_s1_simple,
53     "Study 1")
54
55 ols_interaction_s2 <- get_ols_interaction(ols_anova_s2_simple,
56     "Study 2")
57
58 ols_interaction_results <- bind_rows(ols_interaction_s1,
59     ols_interaction_s2)
60 print(ols_interaction_results)
61
62 write.csv(ols_interaction_results, "ols_interaction_results.csv",
63     row.names = FALSE)
64
65 # -----
66 # 7. Predicted probabilities from ordered logit
67 # Probability of the highest jealousy category
68 # -----
69 newdata_grid <- expand.grid(
70     Attractive_condition = factor(
71         c("Not Attractive", "Attractive"),
72         levels = c("Not Attractive", "Attractive")
73     ),
74     Gender = factor(
75         c("Men", "Women"),
76         levels = c("Men", "Women")
77     )
78 )
79
80 # Study 1
81 pred_probs_s1 <- as.data.frame(
82     predict(ord_model_s1, newdata = newdata_grid, type = "probs")
83 )
84
85 highest_cat_s1 <- tail(colnames(pred_probs_s1), 1)

```

```

80
81 plot_s1 <- cbind(newdata_grid, pred_probs_s1) %>%
82   transmute(
83     Study = "Study 1",
84     Attractive_condition,
85     Gender,
86     Prob_Highest_Jealousy = .data[[highest_cat_s1]]
87   )
88
89 # Study 2
90 pred_probs_s2 <- as.data.frame(
91   predict(ord_model_s2, newdata = newdata_grid, type = "probs")
92 )
93 highest_cat_s2 <- tail(colnames(pred_probs_s2), 1)
94
95 plot_s2 <- cbind(newdata_grid, pred_probs_s2) %>%
96   transmute(
97     Study = "Study 2",
98     Attractive_condition,
99     Gender,
100    Prob_Highest_Jealousy = .data[[highest_cat_s2]]
101  )
102
103 plot_data_ord <- bind_rows(plot_s1, plot_s2)
104
105 print(plot_data_ord)
106 write.csv(plot_data_ord,
107   "ordered_logit_predicted_probabilities.csv", row.names = FALSE)
108
109 # -----
110 # 8. Plot predicted probability of highest jealousy
111 # -----
112 p_ord <- ggplot(
113   plot_data_ord,
114   aes(
115     x = Attractive_condition,
116     y = Prob_Highest_Jealousy,
117     group = Gender,

```

```

117     shape = Gender,
118     linetype = Gender
119   )
120 ) +
121   geom_line(position = position_dodge(width = 0.08)) +
122   geom_point(size = 3, position = position_dodge(width = 0.08)) +
123   facet_wrap(~ Study) +
124   labs(
125     title = "Predicted probability of the highest jealousy
126             category",
127     x = "Rival attractiveness condition",
128     y = "Predicted probability"
129   ) +
130   theme_minimal(base_size = 12)
131 print(p_ord)
132
133 ggsave(
134   filename = "ordered_logit_predicted_probability_plot.png",
135   plot = p_ord,
136   width = 8,
137   height = 5,
138   dpi = 300
139 )
140
141 # -----
142 # 9. Build a comparison table: Ordered logit vs OLS
143 # -----
144 comparison_table <- data.frame(
145   Study = c("Study 1", "Study 2"),
146   OLS_F_value = c(ols_interaction_s1$F_value,
147                   ols_interaction_s2$F_value),
148   OLS_p_value = c(ols_interaction_s1$p_value,
149                   ols_interaction_s2$p_value),
150   OrdLogit_Estimate = c(ord_interaction_s1$Estimate,
151                          ord_interaction_s2$Estimate),
152   OrdLogit_p_value = c(ord_interaction_s1$p_value,
153                          ord_interaction_s2$p_value)

```

6 Extension Results

6.1 Extension 1 Results

Figure 1 shows mean jealousy by rival attractiveness, gender, and study. In Study 1, men’s mean jealousy rises from approximately 2.8 under the non-attractive condition to approximately 3.3 under the attractive condition, while women’s mean jealousy rises from approximately 2.3 to approximately 3.6. The slope is visibly steeper for women, indicating a stronger attractiveness effect among women.

In Study 2, the pattern is weaker. Women’s mean jealousy increases only slightly when the rival is attractive, and men’s mean jealousy appears flat or slightly lower. This indicates that the attractiveness-by-gender interaction is not equally strong across studies.

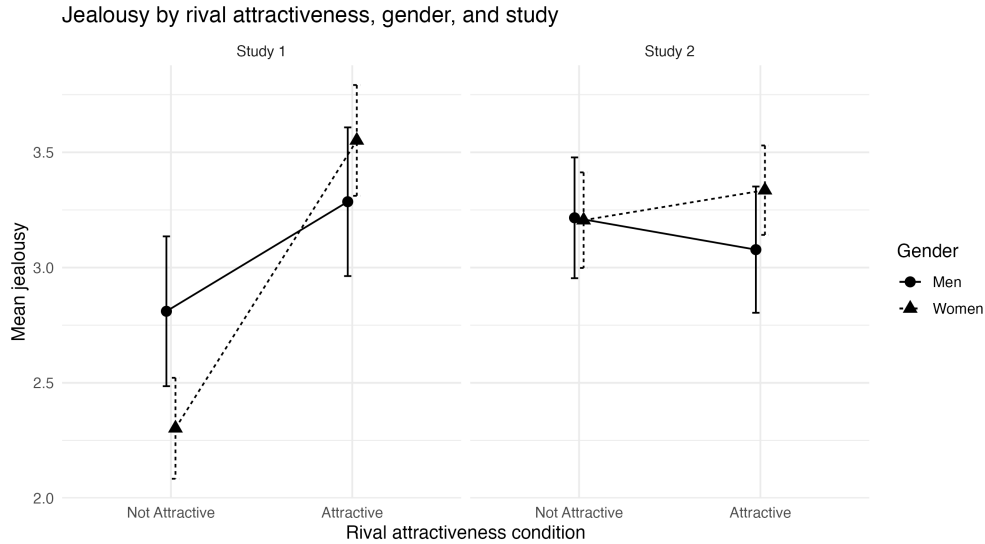


Figure 1: Jealousy by rival attractiveness, gender, and study

Table 3: Pooled Comparison: Three-Way Interaction Effect

Study	Term	Df	Sum Sq	Mean Sq	F-value	p-value
Pooled comparison	Study:Attractive_condition:Gender	2.452		2.452	1.591	0.208

Table 4: Descriptive Statistics of Jealousy by Study, Gender, and Condition

Study	Gender	Attractive Condition	N	Mean	SD	SE	CI Low
Study 1	Men	Not Attractive	58	2.81	1.26	0.166	2.49
Study 1	Men	Attractive	56	3.29	1.23	0.165	2.96
Study 1	Women	Not Attractive	109	2.30	1.17	0.112	2.08
Study 1	Women	Attractive	116	3.55	1.32	0.123	3.31
Study 2	Men	Not Attractive	88	3.22	1.25	0.134	2.95
Study 2	Men	Attractive	90	3.08	1.33	0.140	2.80
Study 2	Women	Not Attractive	141	3.21	1.26	0.106	3.00
Study 2	Women	Attractive	137	3.34	1.16	0.099	3.14

6.2 Extension 2 Results

Figure 2 presents the predicted probability of the highest jealousy category using the ordered logit model. In Study 1, the probability increases substantially when the rival is attractive, particularly for women. In Study 2, the changes are smaller and less consistent.

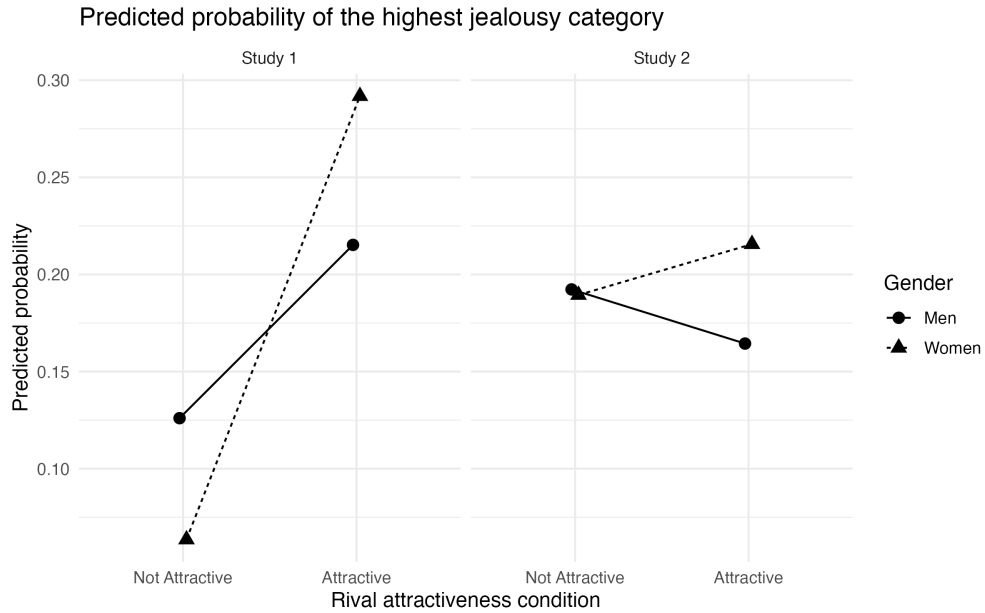


Figure 2: Predicted probability of the highest jealousy category

Table 5: Comparison of OLS and Ordinal Logit Results

Study	OLS F-value	OLS p-value	Ord. Logit Estimate	Ord. Logit p-value
Study 1	7.261	0.0074	0.290	0.0048
Study 2	1.266	0.2612	0.088	0.3044

7 Discussion and Conclusion

This replication project reproduced the original study’s core jealousy models and then extended them in two methodologically relevant ways. The first extension showed that the key attractiveness-by-gender pattern is not equally strong in the two replication studies. Study 1 provides clearer support for the claim, whereas Study 2 shows a weaker pattern.

The second extension showed that the same general conclusion survives re-estimation with an ordered logit model, which is more appropriate for ordinal outcome data. However, it also highlights that the effect is not uniformly large across studies.

Overall, the results suggest that the original finding is not fully robust and is somewhat sample-dependent. At the same time, the consistency across modeling approaches provides some support for the underlying theoretical claim. The main contribution of this report is therefore to evaluate both robustness across samples and robustness across model specifications.

8 References

Pollet TV, Saxton TK. Jealousy as a Function of Rival Characteristics: Two Large Replication Studies and Meta-Analyses Support Gender Differences in Reactions to Rival Attractiveness But Not Dominance. *Pers Soc Psychol Bull.* 2020 Oct;46(10):1428-1443. doi: 10.1177/0146167220904512. Epub 2020 Mar 10. PMID: 32153243; PMCID: PMC7493204.